NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 8: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. While planning using Situation Calculus, determining which predicates (fluents) remain unchanged after an action, without explicitly specifying all unaffected predicates, gives rise to the _____.

A. Frame Problem
B. Successor-state Axiom
C. Possibility Statement
D. Frame Axioms
Correct Answer: (A)

Q2. STRIPS is an alternative representation to the pure situation calculus for planning.

I. Actions are not represented explicitly as part of the world model.
II. Actions are effectively instantaneous.
III. Actions are thought of as operators that syntactically transform world models.

Which of the above statements are correct?

A. Only Statements I and II
B. Statements I, II and III
C. Only Statements II and III
D. None.
Correct Answer: (B)

Q3. ABSTRIPS greatly reduces search space, and is more efficient at solving large problems. It is a kind of _____ planner.

A. State-space
B. Regular
C. Hierarchical
D. Advanced
Correct Answer: (C)

Q4. In a partial-order plan, a causal link from action A to action B for proposition *p* indicates that:

A. A must occur after B and deletes *p*.
B. A achieves *p* for B and protects *p*.
C. A and B are mutually exclusive.
D. *p* remains an open precondition.

Correct Answer: (B)

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Q5. The planning algorithm that breaks up a set of goal predicates into individual subgoals and attempts to solve them individually one after another in a linear order is _____

A. Forward Planning
B. Backward Planning
C. Goal Stack Planning
D. Hierarchical Task Network Planning

Correct Answer: (C)

Q6. Progression and regression are state-space search methods that build a plan as a simple list of steps, i.e., they generate _____ plans; consequently, unlike plan-space planning, they cannot take advantage of problem decomposition.

A. heuristics
B. totally ordered
C. partially ordered
D. uninformed

Correct Answer: (B)

Q7. A planning graph consists of alternating layers of:

A. States and goals.
B. Propositions and actions.
C. Actions and situations.
D. Goals and operators.

Correct Answer: (B)

Q8. Suppose an action S3 threatens a causal link S1 —p→ S2 by deleting p. Which is a valid way to remove the threat?

A. Force S1 and S2 to occur simultaneously.
B. Delete the causal link permanently.
C. Order S3 before S1or after S2.
D. Place S3 between S1 and S2.

Correct Answer: (C)

Q9. Two actions in the same action level of a planning graph are mutex because one action deletes a precondition of the other. This is called:

A. Inconsistent effects.
B. Competing needs.
C. Interference.
D. Inconsistent support.

Correct Answer: (C)

Q10. The Sussman anomaly demonstrates a weakness of non-interleaved planning because:

A. Subgoals can always be solved independently.
B. Solving one subgoal may undo another.
C. Actions cannot have multiple preconditions.
D. Planning graphs cannot represent subgoals.

Correct Answer: (B)

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